

A Cheatsheet for Lie Algebra

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Chapter 1

Lie Algebra

1.1 Basic Concepts

Definition 1.1.1. A vector space L over a field F , with an operation $L \times L \rightarrow L$ $(x, y) \mapsto [xy]$ called the bracket of x, y is called a Lie algebra over F if the following axioms are satisfied :

$$\left\{ \begin{array}{l} \text{the bracket operation is bilinear} \\ [xx] = 0, \forall x \in L \\ [x[yz]] + [y[zx]] + [z[xy]] = 0, \forall x, y, z \in L \end{array} \right.$$

$\rightarrow [x + y, x + y] = [xx] + [xy] + [yx] + [yy]$ by the axiom 1
 $= 0 + [xy] + [yx] + 0$ by the axiom 2 and therefore $[xy] = -[yx]$ and the converse holds given $\text{char} F \neq 2$.

Definition 1.1.2. Two Lie algebras L, L' over F are isomorphic if there exists a vector space isomorphism $\phi : L \rightarrow L'$ satisfying $\phi([xy]) = [\phi(x)\phi(y)]$, $\forall x, y \in L$.

Definition 1.1.3. A subspace K of L is called a subalgebra if $[xy] \in K$, $\forall x, y \in K$.

$\rightarrow \forall 0 \neq x \in L$ it defines a 1-dimensional subalgebra Fx with trivial multiplication by the axiom 2 of the definition.

If V is a finite dimensional vector space over F , denoted by $\text{End}V$, the set of linear transformations $V \rightarrow V$. As a vector space over F , $\text{End}V$ has dimension n^2 where $n = \dim V$, and $\text{End}V$ is a ring relative to the usual product operation. Define a new operation $[xy] = xy - yx$, called the bracket of x, y . With this operation, $\text{End}V$ is a Lie algebra over F .

pf. Axiom 1, 2 are immediate. $[x[yz]] + [y[zx]] + [z[xy]] = [x(yz - zy)] + [y(zx - xz)] + [z(xy - yx)] = [x(yz)] - [x(zy)] + [y(zx)] - [y(xz)] + [z(xy)] - [z(yx)] = x(yz) - (yz)x - x(zy) + (zy)x + y(zx) - (zx)y - y(xz) + (xz)y + z(xy) - (xy)z - z(yx) + (yx)z = 0$ by the

associativity of multiplication operation. Hence the axiom 3 follows and therefore it is a Lie algebra.

Write $\mathfrak{gl}(V)$ for $\text{End}V$ viewed as Lie algebra and call it **the general linear algebra**. Any subalgebra of a Lie algebra $\mathfrak{gl}(V)$ is called a **linear Lie algebra**. Here $\mathfrak{gl}(V)$ can be identified with the set of all $n \times n$ matrices over F , denoted $\mathfrak{gl}(n, F)$.

e.g.

- Let $\dim V = l + 1$. Denote by $\mathfrak{sl}(V) = \mathfrak{sl}(l + 1, F)$ the set of endomorphisms of V having trace zero. Since $\begin{cases} T(xy) = T(yx) \\ T(x + y) = T(x) + T(y) \end{cases}$, $\mathfrak{sl}(V)$ is a subalgebra of $\mathfrak{gl}(V)$, called the **special linear algebra**.
- – when $\dim V = 2l$ with basis $(v_1, \dots, v_{2l})$, denote by $\mathfrak{sp}(V) = \mathfrak{sp}(2l, F)$, the symplectic algebra, which by definition consists of all endomorphisms x of V satisfying $f(x(v), w) = -f(v, x(w))$.
- when $\dim V = 2l + 1$ with basis $(v_1, \dots, v_{2l+1})$, denote by $\mathfrak{o}(V) = \mathfrak{o}(2l + 1, F)$, the orthogonal algebra, which by definition consists of all endomorphisms x of V satisfying $f(x(v), w) = -f(v, x(w))$.
- $\begin{cases} \mathfrak{t}(n, F) \text{ the set of upper triangular matrices } (a_{ij}), a_{ij} = 0, \forall i > j \\ \mathfrak{n}(n, F) \text{ the set of the strictly upper triangular matrices } (a_{ij}), a_{ij} = 0, \forall i \geq j \\ \mathfrak{d}(n, F) \text{ the set of all diagonal matrices} \end{cases} \rightarrow$
 $\mathfrak{t}(n, F) = \mathfrak{d}(n, F) + \mathfrak{n}(n, F)$ with $[\mathfrak{d}(n, F)\mathfrak{n}(n, F)] = \mathfrak{n}(n, F)$.
- Lie algebra of derivation

By an F -algebra, it is a vector space $\mathfrak{U}$ over F endowed with a bilinear operation $\mathfrak{U} \times \mathfrak{U} \rightarrow \mathfrak{U}$. By a derivation of $\mathfrak{U}$, it is a linear map $\delta : \mathfrak{U} \rightarrow \mathfrak{U}$ satisfying the product rule $\delta(ab) = a\delta(b) + \delta(a)b$. Then the collection $\text{Der}\mathfrak{U}$ of all derivation of $\mathfrak{U}$ is a vector subspace of $\text{End}\mathfrak{U}$. Also the bracket of two derivations $[\delta\delta']$ is again derivation, and therefore $\text{Der}\mathfrak{U}$ is a subalgebra of $\mathfrak{gl}(\mathfrak{U})$. Then since a Lie algebra L is an F -algebra in the above sense, $\text{Der}L$ can be defined.

e.g. If $x \in L$, $y \mapsto [xy]$ is an endomorphism of L , denoted adx . Then $\text{adx} \in \text{Der}L$ by the Jacobi identity in the form $[x[yz]] = [[xy]z] + [y[xz]]$. Derivation of this form are called **inner**, all others **outer**. The map $\begin{matrix} L & \rightarrow & \text{Der}L \\ x & \mapsto & \text{adx} \end{matrix}$ is called the **adjoint representation** of L .

If L is an arbitrary finite dimensional vector space over F , we can view L as a Lie algebra by setting $[xy] = 0$, $\forall x, y \in L$. Such an algebra is called **abelian**. If L is any Lie algebra, with basis $x_1, \dots, x_n$, the entire multiplication table of L can be recovered from the structure constants a_{ij}^k s.t. $[x_i x_j] = \sum_{k=1}^n a_{ij}^k x_k$. Then by the axiom 2,3 of the definition gives
$$\begin{cases} a_{ii}^k = 0 = a_{ij}^k + a_{ji}^k \\ \sum_k (a_{ij}^k a_{kl}^m + a_{jl}^k a_{ki}^m + a_{li}^k a_{kj}^m) = 0 \end{cases}.$$

A subspace I of a Lie algebra L is called an **ideal** of L if $\forall x \in L, \forall y \in I, [xy] \in I$. Then $0, L$ are ideals of L trivially. The centre $Z(L) = \{z \in L | [xz] = 0, \forall x \in L\}$ is also an ideal of L . If I, J are two ideals of a Lie algebra L , then
$$\begin{cases} I + J = \{x + y | x \in I, y \in J\} \\ [IJ] = \{\sum x_i y_i | x_i \in I, y_i \in J\} \end{cases}$$
 are ideals.

Definition 1.1.4. If L has no ideals except itself and 0 , and moreover $[LL] \neq 0$, L is called *simple*.

$$\rightarrow L \text{ is simple} \rightarrow Z(L) = 0, L = [LL].$$

e.g. $L = \mathfrak{sl}(2, F)$, $\text{char } F \neq 2$. Take the standard basis of L , $x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $y = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $h =$

$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Then
$$\begin{cases} [xy] = h \\ [hx] = 2x \\ [hy] = -2y \end{cases}$$
. If $I \neq 0$ is an ideal of L , let $ax + by + ch$ be an arbitrary nonzero element of I . Applying adx twice, we get $-2bx \in I$, and applying ady twice, we get $-2ay \in I$. Thus if $a, b \neq 0$, $y \in I$ or $x \in I$ and clearly $I = L$. If $a = b = 0$, $0 \neq ch \in I \rightarrow h \in I$ and $I = L$. So L is simple.

The construction of a quotient algebra L/I is same as the construction of the quotient ring. As vector space L/I is the quotient space, but Lie quotient algebra L/I has an extra structure $[x + I, y + I] = [xy] + I$.

The normaliser of a subalgebra K of L is defined by $N_L(K) = \{x \in L | [xK] \subset K\}$. Then by the Jacobi identity, it is a subalgebra of L . If $K = N_L(K)$, we call K self-normalising.

Definition 1.1.5. Given two Lie algebras L, L' , a linear transformation $\phi : L \rightarrow L'$ is called a *homomorphism* if $\phi([xy]) = [\phi(x)\phi(y)]$, $\forall x, y \in L$. ϕ is called a *monomorphism* if $\ker \phi = 0$, an *epimorphism* if $\text{Im } \phi = L'$, an *isomorphism* if it is both monomorphism and epimorphism.

$$\rightarrow \begin{cases} \ker \phi \text{ is an ideal of } L (\forall x \in \ker \phi, \forall y \in L, \phi([xy]) = [\phi(x)\phi(y)] = 0) \\ \text{Im } \phi \text{ is a subalgebra of } L' \end{cases}.$$

Proposition 1.1.1. 1. If $\phi : L \rightarrow L'$ is a homomorphism of Lie algebras, then $L/\ker \phi \cong \text{Im} \phi$. If I is any ideal of L included in $\ker \phi$, there exists a unique

homomorphism $\psi : L/I \rightarrow L'$ making the diagram

$$\begin{array}{ccc} L & \xrightarrow{\phi} & L' \\ & \searrow \pi & \uparrow \psi \\ & & L/I \end{array} \quad \text{commutes.}$$

2. If I, J are ideals of L s.t. $I \subset J$, J/I is an ideal of L/I and $(L/I)/(J/I)$ is naturally isomorphic to L/J .

3. If I, J are ideals of L , there is a natural isomorphism between $(I + J)/J$ and $I/(I \cap J)$.

pf. by the general results of linear algebras.

Definition 1.1.6. A representation of a Lie algebra L is a homomorphism $\phi : L \rightarrow \mathfrak{gl}(V)$.

e.g. the adjoint representation $\begin{array}{ccc} \text{ad} : L & \rightarrow & \mathfrak{gl}(L) \\ x & \mapsto & \text{adx} \end{array}$. Then $[\text{adxady}](z) = \text{adxady}(z) - \text{adyadx}(z) = \text{adx}([yz]) - \text{ady}([xz]) = [x[yz]] - [y[xz]] = [x[yz]] + [[xz]y] = [[xy]z] = \text{ad}[xy](z)$ so it preserves the bracket. The kernel of ad consists of $x \in L$ for which $\text{adx} = 0$, i.e. for which $[xy] = 0, \forall y \in L$. So $\ker \text{ad} = Z(L)$.

Definition 1.1.7. An automorphism of L is an isomorphism of L onto itself. $\text{Aut} L$ denotes the group of all such.

e.g. If $g \in \text{GL}(V)$ is any invertible endomorphism of V and $gLg^{-1} = L$, then the map $x \mapsto gxg^{-1}$ is an automorphism of L .

Assume $\text{char} F = 0$. Suppose $x \in L$ is an element for which adx is nilpotent, i.e. $(\text{adx})^k = 0, k > 0$. Then the exponential power series for a linear transformation over $\mathbb{C}$ makes sense over F , because it has only finitely many terms; $\exp(\text{adx}) = 1 + \text{adx} + \frac{(\text{adx})^2}{2!} + \cdots + \frac{(\text{ad})^{k-1}}{(k-1)!}$.

CLAIM : If adx is replaced by an arbitrary nilpotent derivation δ of L , then $\exp(\delta) \in \text{Aut} L$.

pf. $\begin{array}{ll} \exp(\text{adx})\exp(\text{ady}) = \exp(\delta(x))\exp(\delta(y)) & \text{by hypothesis} \\ = (\sum_{i=0}^{k-1} \frac{\delta^i x}{i!})(\sum_{j=0}^{k-1} \frac{\delta^j y}{j!}) & = \sum_{n=0}^{2k-2} (\sum_{i=0}^n \frac{\delta^i x}{i!})(\frac{\delta^{n-i} y}{(n-i)!}) \\ & = \sum_{n=0}^{2k-2} \frac{\delta^n(xy)}{n!} \quad \text{by Leibniz rule} \\ & = \sum_{n=0}^{k-1} \frac{\delta^n(xy)}{n!} \quad \text{since } \delta^k = 0 \\ & = \exp(\delta(xy)) \end{array}$

is invertible since its inverse is $1 - \eta + \eta^2 = \cdots \pm \eta^{k-1}$, $\exp \delta = 1 + \eta$.

An automorphism of the form $\exp(\text{adx})$ where adx is nilpotent is called **inner**. The subgroup of $\text{Aut}L$ generated by these inner automorphisms is denoted $\text{Int}L$ and it is a normal subgroup; $\forall \phi \in \text{Aut}L, \forall x \in L, \phi(\text{adx})\phi^{-1} = \text{ad}\phi(x)$ and therefore $\phi(\exp \text{adx})\phi^{-1} = \exp(\text{ad}\phi(x))$.

e.g. Let $L = \mathfrak{sl}(2, F)$ with standard basis (x, y, h) . Define $\sigma = \exp \text{adx} \cdot \exp \text{ad}(-y) \exp \text{adx} \in$

$\text{Int}L$. Then $\begin{cases} \sigma(x) = -y \\ \sigma(y) = -x \\ \sigma(h) = -h \end{cases}$ with the order of σ 2.

→ if $L \subset \mathfrak{gl}(V)$ is an arbitrary linear Lie algebra with $\text{char } F = 0$, $x \in L$ is nilpotent, then

$$(\exp x)y(\exp x)^{-1} = \exp \text{adx}(y), \quad \forall y \in L$$

pf. $\text{adx} = \lambda_x + \rho_{-x}$ where λ_x, ρ_x denote the left and right multiplication by x in the ring $\text{End}V$. Then $\exp \text{adx} = \exp(\lambda_x + \rho_{-x}) = \exp \lambda_x \cdot \exp \rho_{-x} = \lambda_{\exp x} \cdot \rho_{\exp(-x)}$ so the equation follows.

Definition 1.1.8. Define a sequence of ideals of L called the derived series by $L^{(0)} = L, L^{(1)} = [LL], L^{(2)} = [L^{(1)}L^{(1)}], \dots, L^{(i)} = [L^{(i-1)}L^{(i-1)}]$. Call L solvable if $L^{(n)} = 0$ for some n .

e.g. $L = \mathfrak{t}(n, F)$ is solvable.

pf. The basis for $\mathfrak{t}(n, F)$ consists of the matrix units e_{ij} , $i \leq j$ and the dimension is $\frac{n(n+1)}{2}$. Then $[e_{ii}, e_{il}] = e_{il}$, $i < l$, thus $\mathfrak{n}(n, F) \subset [LL]$, where $\mathfrak{n}(n, F)$ is the subalgebra of upper triangular nilpotent matrices. Since $\mathfrak{t}(n, F) = \mathfrak{d}(n, F) + \mathfrak{n}(n, F)$, with $\mathfrak{d}(n, F)$ abelian, $\mathfrak{n}(n, F)$ is equal to the derived algebra of L . In $\mathfrak{n}(n, F)$, we have a natural notion of level for e_{ij} , namely $j - i$. Assume $i < j, k < l$. Wlog assume $i \neq l$.

Then $[e_{ij}e_{kl}] = \begin{cases} e_{il} & j = k \\ 0 & \text{otherwise} \end{cases}$. Each e_{il} is commutator of two matrices whose

levels add upto that of e_{il} . thus $L^{(2)}$ is spanned by those e_{ij} of level ≥ 2 , $L^{(i)}$ by those of level $\geq 2^{i-1}$. Then it is clear that $L^{(i)} = 0$ whenever $2^{i-1} > n - 1$. The claim follows.

Proposition 1.1.2. Let L be a Lie algebra.

1. If L is solvable, then so are all subalgebras and homomorphic images of L .
2. If I is a solvable ideal of L s.t. L/I is solvable, then L itself is solvable.
3. If I, J are solvable ideals of L , then so is $I + J$.

pf.

- By the definition, if K is a subalgebra of L , $K^{(i)} \subset L^{(i)}$. Similarly, if $\phi: L \rightarrow M$ is an epimorphism, $\phi(L^{(i)}) = M^{(i)}$ gives the stmt.

- Say $(L/I)^{(n)} = 0$. Then by 1 the canonical homomorphism $\pi : L \rightarrow L/I$ has $\pi(L^{(n)}) = 0$ i.e. $L^{(n)} \subset I = \ker \pi$. If $I^{(m)} = 0$, then $(L^{(i)})^{(j)} = L^{(i+j)}$ implies $L^{(n+m)} = 0$.
- An isomorphism between $(I+J)/J$ and $I/(I \cap J)$ exists. As a homomorphic image of I , $I/(I \cap J)$ is solvable, and therefore its isomorphic image $(I+J)/J$ is solvable. Then so is $I+J$ by 2.

→ let S be a maximal solvable ideal of the Lie algebra L . If I is any solvable ideal of L , by 3, $S+I = S$ by maximality or $I \subset S$. Thus there exists the unique maximal solvable ideal, called **the radical** of L , denoted $\text{Rad}L$. If $\text{Rad}L = 0$, L is called **semisimple**.

Definition 1.1.9. Define a sequence of ideals of L called the descending central series by $L^0 = L, L^1 = [LL], L^2 = [LL^1], \dots, L^i = [LL^{i-1}]$. L is called nilpotent if $L^n = 0$ for some n .

→ $L^{(i)} \subset L^i, \forall i$, and therefore nilpotent algebras are solvable, but not vice versa.

Proposition 1.1.3. Let L be a Lie algebra.

1. If L is nilpotent, then so are all subalgebras and homomorphic images of L
2. If $L/Z(L)$ is nilpotent, then so is L
3. If L is nilpotent and nonzero, $Z(L) \neq 0$.

pf.

- By the definition, for any subalgebra K of L , $K^i \subset L^i$. Similarly, if $\phi : L \rightarrow M$ is an epimorphism, $\phi(L^i) = M^i$ gives the stmt.
- If $L^n \subset Z(L)$, $L^{n+1} = [LL^n] \subset [LZ(L)] = 0$.
- The last nonzero term of the descending central series is central.

→ $\exists n$ s.t. $\text{adx}_1, \dots, \text{adx}_n(y) = 0, \forall x_i, y \in L$. If L is any Lie algebra and $x \in L$, we call x **ad-nilpotent** if adx is a nilpotent endomorphism. If L is nilpotent, then all elements of L are ad-nilpotent.

Lemma 1.1.1. Let $x \in \mathfrak{gl}(V)$ be a nilpotent endomorphism. Then adx is also nilpotent.

pf. We may associate to x two endomorphisms of $\text{End}V$; the left and right translation $\begin{cases} \lambda_x(y) = xy \\ \rho_x(y) = yx \end{cases}$, which are nilpotent because x is. Moreover, λ_x, ρ_x commute. In any ring including $\text{End}(V)$, the sum or difference of two commuting nilpotents is again nilpotent, so $\text{adx} = \lambda_x - \rho_x$ is nilpotent.

Theorem 1.1.1. *Let L be a subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. If L consists of nilpotent endomorphisms and $V \neq 0$, $\exists 0 \neq v \in V$ for which $Lv = 0$.*

pf. When $\dim L = 0$ or 1 , the theorem holds trivially.

When $\dim L > 1$, suppose $K \neq L$ be any subalgebra of L . Then by the previous lemma, K acts as a Lie algebra of nilpotent linear transformations on the vector space L , hence also on the vector space L/K . Because $\dim K < \dim L$, by the inductive hypothesis, there exists a vector $x \in L/K$ s.t. $x + K \neq K$ s.t. it is killed by the image of K in $\mathfrak{gl}(L/K)$. This means $[yx] \in K$, $\forall y \in K, x \notin K$, i.e. K properly included in $N_L(K)$. Take K to be a maximal proper subalgebra of L . Then $N_L(K) = L$, i.e. K is an ideal of L . If $\dim L/K$ were greater than 1 , then the inverse image in L of a 1 -dimensional subalgebra of L/K would be a proper subalgebra properly containing K , a contradiction. Thus K has codimension 1 . Thus we can write $L = K + Fz$, where $z \in L \setminus K$. By induction, $W = \{v \in V | Kv = 0\}$ is nonzero. Since K is an ideal, W is stable under L ($\begin{cases} \forall x \in L, \\ \forall y \in K \end{cases}$, if $w \in W$, $yxw = xyw - [xy]w = 0$). Choose $z \in L \setminus K$ as above, so the nilpotent endomorphism z has an eigenvector, i.e. $\exists 0 \neq v \in W$ for which $zv = 0$. Then $Lv = 0$ as desired.

Theorem 1.1.2 (Engel). *If all elements of L are ad-nilpotent, then L is nilpotent.*

pf. Given a Lie algebra L , all of whose elements are ad-nilpotent, $\text{ad}L \subset \mathfrak{gl}(L)$ satisfies the hypothesis of the previous theorem. Thus there exists $x \neq 0$ in L for which $[Lx] = 0$, i.e. $Z(L) \neq 0$. Now $L/Z(L)$ evidently consists of ad-nilpotent elements and has smaller dimension than L . By the inductive hypothesis, $L/Z(L)$ is nilpotent. By the previous proposition 2, L is nilpotent as well.

Definition 1.1.10. *If V is a finite dimensional vector space, a flag in V is a chain of subspaces $0 = V_0 \subset V_1 \subset \dots \subset V_n = V$, $\dim V_i = i$.*

$\rightarrow$ if $x \in \text{End}V$, x is said to stabilises this flag provided $xV_i \subset V_i$, $\forall i$.

Corollary 1.1.1. *Under the hypotheses of the theorem, there exists a flag (V_i) in V stable under L , with $xV_i \subset V_{i-1}$, $\forall i$. In other words, there exists a basis of V relative to which the matrices of L are all in $\mathfrak{n}(n, F)$.*

pf. Begin with any nonzero $v \in V$ killed by L , the existence of which is assured by the theorem. Set $V_1 = Fv$. Let $W = V/V_1$ and the induced action of L on W is also by nilpotent endomorphisms. By the inductive hypotheses, W has a flag stabilised by L , whose inverse image in V does the trick.

Lemma 1.1.2. *Let L be nilpotent, K be an ideal of L . Then if $K \neq 0$, $K \cap Z(L) \neq 0$.*

pf. L acts on K via the adjoint representation, so by the previous theorem, $\exists 0 \neq x \in K$ killed by L , i.e. $[Lx] = 0 \rightarrow x \in K \cap Z(L)$.

1.2 Semisimple Lie Algebras

Theorem 1.2.1. *Let L be a solvable subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. If $V \neq 0$, then V contains a common eigenvector for all the endomorphisms in L .*

pf. When $\dim L = 0$, the theorem trivially holds. When $\dim L > 0$, L is solvable of positive dimension, so L properly includes $[LL]$. Since $L/[LL]$ is abelian, any subspace is an ideal. Take a subspace of codimension 1, then its inverse image K is an ideal of codimension 1 in L . Then by the inductive hypothesis $\exists v \in V$ for K a common eigenvector, i.e. $\forall x \in K, xv = \lambda(x)v$. Fix this linear function $\lambda : K \rightarrow F$, and denote by W the subspace $\{w \in V | xw = \lambda(x)w, \forall x \in K\}$. Then $W \neq 0$.

- **L leaves W invariant**

pf. Fix $w \in W, x \in L$. Let n be the smallest positive integer s.t. $w, xw, \dots, x^n w$ are L.D. Let W_i be the subspace of V spanned by $w, xw, \dots, x^{i-1}w$ so that $\dim W_n = n, W_n = W_{n+i}, \forall i \geq 0$ and x maps W_n into W_n . Then $\forall y \in K$ leaves each W_i invariant. Relative to the basis $w, xw, \dots, x^{n-1}w$ of W_n , $yx^i w = yxx^{i-1}w = xyx^{i-1}w - [xy]x^{i-1}w = x\lambda(y)x^{i-1}w + xw' - [xy]x^{i-1}w$ where $w' \in W_{i-1}$ by the inductive hypothesis and since x maps W_{i-1} into W_i , $yx^i w \equiv \lambda(y)x^i w \pmod{W_i}$. Then $T_{W_n}(y) = n\lambda(y)$. In particular, it is true for elements of K of the special form $[xy]$. But x, y both stabilise W_n , so $[xy]$ acts on W_n as the commutator of two endomorphisms of W_n and therefore its trace is 0. So $n\lambda([xy]) = 0$. Since $\text{char } F = 0$, $\lambda([xy]) = 0$ and L stabilises W .

Then write $L = K + Fz$. Since F is algebraically closed, we can find an eigenvector $v_0 \in W$ of z for some eigenvalue. Then v_0 is a common eigenvector for L and λ can be extended to a linear function on L s.t. $xv_0 = \lambda(x)v_0, x \in L$.

Corollary 1.2.1 (Lie's Theorem). *Let L be a solvable subalgebra of $\mathfrak{gl}(V)$, $\dim V = n < \infty$. Then L stabilises some flag in V , i.e. the matrices of L relative to a suitable basis of V are upper triangular.*

pf. by the induction on $\dim V$ in the use of the previous theorem.
 $\rightarrow$ let L be any solvable Lie algebra, $\phi : L \rightarrow \mathfrak{gl}(V)$ a finite dimensional representation of L . Then $\phi(L)$ is solvable by the previous proposition, hence stabilises a flag by the previous corollary.

Corollary 1.2.2. *Let L be solvable. Then there exists a chain of ideals of L , $0 = L_0 \subset \cdots \subset L_n = L$ s.t. $\dim L_i = i$.*

Corollary 1.2.3. *Let L be solvable. Then $x \in [LL]$ implies $\text{ad}_L x$ is nilpotent. In particular, $[LL]$ is nilpotent.*

pf. find a flag of ideals by the previous corollary. Relative to a basis $(x_1, \dots, x_n)$ of L for which $(x_1, \dots, x_i)$ spans L_i , the matrices of $\text{ad} L$ lie in $\mathfrak{t}(n, F)$. Therefore the matrices of $[\text{ad} L \text{ad} L] = \text{ad}_L [LL]$ lie in $\mathfrak{n}(n, F)$, the derived algebra of $\mathfrak{t}(n, F)$. Thus $\text{ad}_L x$ is nilpotent for $x \in [LL]$. Then $\text{ad}_{[LL]} x$ is nilpotent and so is $[LL]$ by Engel's theorem.

Call $x \in \text{End} V$ **semisimple** if the roots of its minimal polynomial over F are all distinct. Equivalently, x is semisimple iff x is diagonalisable, given F being algebraically closed. If x is semisimple and maps a subspace W of V into itself, then the restriction of x to W is semisimple.

Proposition 1.2.1. *Let V be a finite dimensional vector space over F , $x \in \text{End} V$.*

1. *There exists unique $x_s, x_n \in \text{End} V$ satisfying the conditions : $x = x_s + x_n$, x_s is semisimple, x_n is nilpotent and x_s, x_n commute.*
2. *There exists polynomials $p(T), q(T)$ in one indeterminate, without constant term, s.t. $x_s = p(x)$, $x_n = q(x)$. In particular, x_s, x_n commute with any endomorphism commuting with x .*
3. *If $A \subset B \subset V$ are subspaces and x maps B into A , then x_s, x_n also maps B into A .*

pf.

• 2

Let $a_1, \dots, a_k$ be the distinct eigenvalues of x with multiplicities $m_1, \dots, m_k$. Then the characteristic polynomial is $\Pi(T - a_i)^{m_i}$. If $V_i = \ker(x - a_i 1)^{m_i}$, then V is the direct sum of the subspaces $V_1, \dots, V_k$, each stable under x . On V_i , x clearly has characteristic polynomial $(T - a_i)^{m_i}$. Then by the CRT, we have a polynomial $p(T) \equiv \begin{cases} a_i & \text{mod } (T - a_i)^{m_i} \\ 0 & \text{mod } T \end{cases}$. Set $q(T) = T - p(T)$. Then both

$p(T), q(T)$ have no constant term. Set $\begin{cases} x_s = p(x) \\ x_n = q(x) \end{cases}$. Since they are polynomials in x , x_s, x_n commute with each other as well as all endomorphisms which commute with x .

- 3

x_s, x_n stabilise all subspaces of V stabilised by x , in particular the V_i . The congruence $p(T) \equiv a_i \pmod{(T - a_i)^{m_i}}$ shows the restriction of $x_s - a_i 1$ to V_i is zero for all i , hence x_s acts diagonally on V_i with single eigenvalue a_i . By definition, $x_n = x - x_s$, so x_n is nilpotent trivially. Since $p(T), q(T)$ have no constant term, 3 follows.

- 1

Let $x = s + n$ be another such decomposition, so $x_s - s = n - x_n$. By 2, all endomorphisms commute. Sums of commuting semisimple and nilpotent endomorphisms are semisimple and nilpotent respectively. Thus $s = x_s, n = x_n$ since only 0 can be both semisimple and nilpotent.

→ the decomposition $x = x_s + x_n$ is called **the additive Jordan-Chevalley decomposition of x** . x_s, x_n are called the semisimple part and the nilpotent part of x respectively.

Consider the adjoint representation of the Lie algebra $\mathfrak{gl}(V)$, V finite dimensional. If $x \in \mathfrak{gl}(V)$ is nilpotent, then so is adx by the previous lemma.

CLAIM : If x is semisimple, so is adx .

pf. Choose a basis $(v_1, \dots, v_n)$ of V relative to which x has matrix $\text{diag}(a_1, \dots, a_n)$. Let $\{e_{ij}\}$ be the standard basis of $\mathfrak{gl}(V)$ relative to $(v_1, \dots, v_n)$ s.t. $e_{ij}(v_k) = \delta_{jk} v_i$. Then $\text{adx}(e_{ij}) = (a_i - a_j)e_{ij}$, so adx has diagonal matrix, relative to the chosen basis of $\mathfrak{gl}(V)$.

Lemma 1.2.1. *Let $x \in \text{End} V$, $x = x_s + x_n$ be its Jordan decomposition. Then $\text{adx} = \text{adx}_s + \text{adx}_n$ is the Jordan decomposition of adx .*

pf. adx_s is semisimple and adx_n is nilpotent, and they commute, since $[\text{adx}_s \text{adx}_n] = \text{ad}[x_s x_n] = 0$. Then by the previous proposition 1, the lemma follows.

Lemma 1.2.2. *Let $\mathfrak{U}$ be a finite dimensional F -algebra. Then $\text{Der} \mathfrak{U}$ contains the semisimple and nilpotent parts of all its elements.*

pf. If $\delta \in \text{Der} \mathfrak{U}$, let $\sigma, \nu \in \text{End} \mathfrak{U}$ be its semisimple and nilpotent parts respectively. If $a \in F$, set $\mathfrak{U}_a = \{x \in \mathfrak{U} | (\delta - a 1)^k x = 0, \exists k\}$. Then $\mathfrak{U}$ is the direct sum of those $\mathfrak{U}_a$ for which a is an eigenvalue of δ and σ acts on $\mathfrak{U}_a$ as scalar multiplication by a . Then $\forall a, b \in F, \mathfrak{U}_a \mathfrak{U}_b \subset \mathfrak{U}_{a+b}$ since $(\delta - (a+b)1)^n(xy) = \sum_{i=0}^n ((\delta - a1)^{n-i} x) \cdot ((\delta - b1)^i y)$, $x, y \in \mathfrak{U}$. Then if $x \in \mathfrak{U}_a, y \in \mathfrak{U}_b$, $\sigma(xy) = (a+b)xy$, since $xy \in \mathfrak{U}_{a+b}$. On the other hand, $(\sigma x)y + x(\sigma y) = (a+b)xy$. By the directness of the sum $\mathfrak{U} = \coprod \mathfrak{U}_a$, σ is a derivation. Since $\nu = \delta - \sigma$ and both $\delta, \sigma \in \text{Der} \mathfrak{U}$, so is ν .

Lemma 1.2.3. *Let $A \subset B$ be two subspaces of $\mathfrak{gl}(V)$, $\dim V < \infty$. Set $M = \{x \in \mathfrak{gl}(V) \mid [xB] \subset A\}$. Suppose $x \in M$ satisfies $T(xy) = 0$, $\forall y \in M$. Then x is nilpotent.*

pf. Let $x = s + n$ be the Jordan decomposition of x . Fix a basis $v_1, \dots, v_m$ of V relative to which s has matrix $\text{diag}(a_1, \dots, a_m)$. Let E be the vector subspace of F over the prime field $\mathbb{Q}$ spanned by the eigenvalues $a_1, \dots, a_m$. Since E has finite dimension over $\mathbb{Q}$, we have the dual space E^* consisting of linear functions $f : E \rightarrow \mathbb{Q}$. Given f , let y be the element of $\mathfrak{gl}(V)$ whose matrix relative to the given basis is $\text{diag}(f(a_1), \dots, f(a_m))$. If $\{e_{ij}\}$ is the corresponding basis of $\mathfrak{gl}(V)$, $\begin{cases} \text{ads}(e_{ij}) = (a_i - a_j)e_{ij} \\ \text{ady}(e_{ij}) = (f(a_i) - f(a_j))e_{ij} \end{cases}$. Let $r(T) \in F[T]$ be a polynomial without constant term satisfying $r(a_i - a_j) = f(a_i) - f(a_j)$ for all pairs i, j . The existence follows from the Lagrange interpolation. Evidently $\text{ady} = r(\text{ads})$. Then ads is the semisimple part of adx by the lemma 1.2.2., so it can be written as a polynomial in adx without constant term by the proposition 0.2.1. Therefore ady is also a polynomial in adx without constant term. By the hypothesis, adx maps B into A so $\text{ady}(B) \subset A$, i.e. $y \in M$. Then since $y = \text{diag}(f(a_i))$ and only diagonal part of x contributes to $T(xy)$, $T(xy) = 0$ and $\sum a_i f(a_i) = 0$. The LHS is a $\mathbb{Q}$ -linear combination of elements of E , and applying f gives $\sum f(a_i)^2 = 0$, but the numbers $f(a_i) \in \mathbb{Q}$ so all of them are forced to be 0. Thus f must be identically 0, i.e. E^* is zero, and therefore so is E .

CLAIM : If x, y, z are endomorphisms of a finite dimensional vector spaces,
 $T([xy]z) = T(x[yz])$.

pf. $T([xy]z) = T(xyz - yxz) = T(xyz - xzy) = T(x[yz])$ since $T(y(xz)) = T((xz)y)$.

Theorem 1.2.2 (Cartan's Criterion). *Let L be a subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. Suppose $T(xy) = 0$, $\forall x \in [LL]$, $\forall y \in L$. Then L is solvable.*

pf. Let $A = [LL]$, $B = L$ so $M = \{x \in \mathfrak{gl}(V) \mid [xL] \subset [LL]\}$. Then $L \subset M$. By the hypothesis $T(xy) = 0$, $\forall x \in [LL]$, $\forall y \in L$. If $[xy]$ is a typical generator of $[LL]$ and if $z \in M$, then by the previous claim, $T([xy]z) = T(x[yz]) = T([yz]x)$. Then by the definition of M , $[yz] \in [LL]$, so it is 0 by hypothesis. So by the previous lemma, $x \in [LL]$ are nilpotent so the $\text{ad}_{[LL]}x$ is nilpotent. Thus $[LL]$ is nilpotent by Engel's theorem and $[LL]$ is solvable. Since $L/[LL]$ is abelian, the theorem follows by the previous proposition 1.1.3..

Corollary 1.2.4. *Let L be a Lie algebra s.t. $T(\text{adxady}) = 0$, $\forall x \in [LL]$, $\forall y \in L$. Then L is solvable.*

pf. By the previous theorem, the adjoint representation of L gives $\text{ad}L$ solvable. Since $\ker \text{ad} = Z(L)$ is solvable given the centre being abelian, L itself is solvable by the previous proposition 1.1.3.

Definition 1.2.1. Let L be any Lie algebra. If $x, y \in L$, define $\kappa(x, y) = T(\text{adx} \text{ady})$.

$\rightarrow \kappa$ is associative, in the sense that $\kappa([xy], z) = \kappa(x, [yz])$.

Lemma 1.2.4. Let I be an ideal of L . If κ is the killing form of L and κ_I the killing form of I , then $\kappa_I = \kappa|_{I \times I}$.

pf. If $x, y \in I$, $(\text{adx})(\text{ady})$ is an endomorphism of L , mapping L into I , so its trace $\kappa(x, y)$ coincides with the trace $\kappa_I(x, y)$ of $(\text{adx})(\text{ady})|_I = (\text{ad}_I x)(\text{ad}_I y)$.

A symmetric bilinear form $\beta(x, y)$ is called **nondegenerate** if its radical S is 0, where $S = \{x \in L | \beta(x, y) = 0, \forall y \in L\}$. Because the killing form is associative, S is an ideal of L .

We can test nondegeneracy as follows; fix a basis $x_1, \dots, x_n$ of L . Then κ is nondegenerate iff the $n \times n$ matrix whose i, j entry is $\kappa(x_i, x_j)$ has nonzero determinant.

e.g. $L = \mathfrak{sl}(2, F)$ with the standard basis (x, h, y) gives $\text{adh} = \text{diag}(2, 0, -2)$, $\text{adx} = \begin{bmatrix} 0 & -2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, $\text{ady} = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}$ and therefore $\kappa = \begin{bmatrix} 0 & 0 & 4 \\ 0 & 8 & 0 \\ 4 & 0 & 0 \end{bmatrix}$ with determinant -128, and κ is nondegenerate, given $\text{char} F \neq 2$.

A Lie algebra L is called semisimple in case $\text{Rad} L = 0 \Leftrightarrow L$ has no nonzero abelian ideals. Conversely, the radical includes such an ideal of L , the last nonzero term in the derived series of $\text{Rad} L$.

Theorem 1.2.3. Let L be a Lie algebra. Then L is semisimple iff its killing form is nondegenerate.

pf.

- Suppose $\text{Rad} L = 0$. Let S be the radical of κ . Then $T(\text{adx} \text{ady}) = 0, \forall x \in S, y \in L$. By the Cartan's Criterion, $\text{ad}_L S$ is solvable, hence S is solvable. Since S is an ideal of L , $S \subset \text{Rad} L = 0$, κ is nondegenerate.
- Suppose $S = 0$ and for any abelian ideal I of L , $x \in I, y \in L$. Then $\text{adx} \text{ady}$ maps $L \rightarrow L \rightarrow I$ and $(\text{adx} \text{ady})^2$ maps L into $[II] = 0$, i.e. $\text{adx} \text{ady}$ is nilpotent, hence $0 = T(\text{adx} \text{ady}) = \kappa(x, y)$, so $I \subset S = 0$. So L is semisimple.

$\rightarrow S \subset \text{Rad} L$, but not vice versa.

pf. Let L be a 3 dimensional vector space with bases (x, y, z) s.t. $[xy] = z, [xz] = y, [yz] = 0$. Then $\text{adx} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $\text{ady} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $\text{adz} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $\kappa =$

$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and therefore $S = \text{Rad}\kappa = (y, z)$. Since $[LL] = \text{span}(y, z)$ and $[[LL], [LL]] = 0$, L is solvable, hence $\text{Rad}L = L$. Then $\text{Rad}L \not\subset S$.

Definition 1.2.2. A Lie algebra L is said to be the direct sum of ideals $I_1, \dots, I_t$ provided $L = I_1 + \dots + I_t$ and written as $L = I_1 \oplus \dots \oplus I_t$.

$$\rightarrow [I_i I_j] \subset I_i \cap I_j = 0, \forall i \neq j.$$

Theorem 1.2.4. Let L be semisimple. Then there exist ideals $L_1, \dots, L_t$ of L which are simple, s.t. $L = L_1 \oplus \dots \oplus L_t$. Every simple ideal of L coincides with one of the L_u . Moreover, the killing form of L_i is the restriction of κ to $L_i \times L_i$.

pf.

- **For an arbitrary ideal I , $L = I \oplus I^\perp$**

Let I be an arbitrary ideal of L . Then $I^\perp = \{x \in L \mid \kappa(x, y) = 0, \forall y \in I\}$ is also an ideal by the associativity of κ . By Cartan's criterion, the ideal $I \cap I^\perp$ of L is solvable and therefore $L = I \oplus I^\perp$ given $\dim I + \dim I^\perp = \dim L$.

- existence

If L has no nonzero proper ideal, then L is simple and there is nothing to prove.

Otherwise, let L_1 be a minimal nonzero ideal. Then by the stmt above, $L = L_1 \oplus L_1^\perp$. In particular, any ideal of L_1 is also an ideal of L , so L_1 is semisimple, hence simple by minimality. For the same reason, $L_1^\perp$ is semisimple. Then by inductive hypothesis, it splits into a direct sum of simple ideals, which are also ideals of L . So the decomposition of L follows.

- uniqueness

If I is any simple ideal of L , $[IL]$ is also an ideal of I , nonzero, given $Z(L) = 0$. Thus $[IL] = I$. On the other hand, $[IL] = [IL_1] \oplus \dots \oplus [IL_t]$, so all but one summand must be 0. Say $[IL_1] = I$. Then $I \subset L_1$ and $I = L_1$ given L_1 simple.

Corollary 1.2.5. If L is semisimple, then $L = [LL]$ and all ideals and homomorphic images of L are semisimple. Moreover, each ideal of L is a sum of certain simple ideals of L .

$\text{ad}L$ is an ideal in $\text{Der}L$.

pf. $[\delta, \text{ad}x] = \text{ad}(\delta x), \forall x \in L, \delta \in \text{Der}L$.

Theorem 1.2.5. *If L is semisimple, then $\text{ad}L = \text{Der}L$, i.e. every derivation of L is inner.*

pf. Since L is semisimple, $Z(L) = 0$. Therefore $L \rightarrow \text{ad}L$ is an isomorphism of Lie algebras. In particular, $M = \text{ad}L$ itself has nondegenerate killing form by the previous theorem 1.2.3. If $D = \text{Der}L$, $[DM] \subset M$ by the previous stmt. Thus κ_M is the restriction to $M \times M$ of the killing form κ_D of D . In particular, if $I = M^\perp$ is the subspace of D orthogonal to M under κ_D , then the nondegeneracy of κ_M forces $I \cap M = 0$. Since both I, M are ideals of D , $[IM] = 0$. If $\delta \in I$, $\text{ad}(\delta x) = 0$, $\forall x \in L$ by the previous stmt so $\delta x = 0$, given ad is one to one and $\delta = 0$. Thus $I = 0$, $\text{Der}L = M = \text{ad}L$.

$\rightarrow$ since $\text{Der}L$ coincides with $\text{ad}L$, each $x \in L$ determines unique elements $s, n \in L$ s.t. $\text{ad}x = \text{ad}s + \text{ad}n$ the Jordan decomposition of $\text{ad}x$ in $\text{End}L$, given $L \rightarrow \text{ad}L$ being one to one. This means $x = s + n$ with $[sn] = 0$, $\begin{cases} s \text{ ad-semisimple} \\ n \text{ ad-nilpotent} \end{cases}$. Write $s = s_x, n = n_x$ and call these the semisimple and nilpotent parts of x .

Definition 1.2.3. *Let L be a Lie algebra. A vector space V , endowed with an operation $L \times V \rightarrow V$ is called an L -module if the following conditions are satisfied;*

$$\begin{cases} (ax + by)v = a(xv) + b(yv) & x, y \in L \\ x(av + bw) = a(xv) + b(xw) & v, w \in V \\ [xy]v = xyv - yxv & a, b \in F \end{cases}$$

e.g. for $\phi : L \rightarrow \mathfrak{gl}(V)$, a representation of L , V may be viewed as an L -module via the action $xv = \phi(x)(v)$. Conversely, given an L -module V , this equation defines a representation $\phi : L \rightarrow \mathfrak{gl}(V)$.

Definition 1.2.4. *A homomorphism of L -modules is a linear map $\phi : V \rightarrow W$ s.t. $\phi(xv) = x\phi(v)$.*

$\rightarrow \ker \phi$ is an L -submodule of V .

An L -module V is called **irreducible** if it has precisely two L -submodules (itself and 0). V is called **completely reducible** if V is a direct sum of irreducible L -submodules, or equivalently if each L -submodule W of V has a complement W' .

Given a representation $\phi : L \rightarrow \mathfrak{gl}(V)$, the associative algebra generated by $\phi(L)$ in $\text{End}V$ leaves invariant precisely the same subspaces as L . Thus all the usual results for modules over associative rings hold for L .

Schur's Lemma: **Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be irreducible. The only endomorphisms of V commuting with all $\phi(x)$, $\forall x \in L$ are scalars.**

- Let V be an L -module. The dual vector space V^* becomes an L -module called **contragredient** if we define $\forall f \in V^*, \forall v \in V, \forall x \in L, (xf)(v) = -f(xv)$.

$$\begin{aligned}
& ([xy]f)(v) = -f([xy]v) \quad \text{by definition of operation} \\
& \quad = -f(xyv - yxv) \quad \text{by the definition of bracket} \\
\text{pf. } & = -f(xyv) + f(yxv) \quad \text{by the linearity} \quad \text{so it is a module.} \\
& = (xf)(yv) - (yf)(xv) \quad = -(yxf)(v) + (xyf)(v) \\
& = ((xy - yx)f)(v)
\end{aligned}$$

- If V, W are L -modules, let $V \otimes W$ be the tensor product over F of the underlying vector spaces. Then $x(v \otimes w) = xv \otimes w + v \otimes xw$, hence it is a module.
 pf. $[xy](v \otimes w) = [xy]v \otimes w + v \otimes [xy]w = (xyv - yxv) \otimes w + v \otimes (xyw - yxw) = (xyv \otimes w + v \otimes xyw) - (yxv \otimes w + v \otimes yxw) = (xy - yx)(v \otimes w)$, hence it is a module.

e.g. Given a vector space V over F , $V^* \otimes V \rightarrow \text{End} V$
 $f \otimes v \mapsto (w \mapsto f(w)v)$. If in addition V is an L -module, then $V^* \otimes V$ is an L -module in the way described above. Thus $\text{End} V$ can also be viewed as an L -module, s.t. $(xf)(v) = xf(v) - f(xv)$, $\forall x \in L, f \in \text{End} V, v \in V$.

Let L be semisimple and $\phi : L \rightarrow \mathfrak{gl}(V)$ be faithful representation of L . Define a symmetric bilinear form $\beta(x, y) = T(\phi(x)\phi(y))$ on L . Then β is associative, so its radical S is an ideal of L . Moreover β is nondegenerate (by the Cartan's criterion, $\phi(S) \cong S$ is solvable, so $S = 0$). Now let L be semisimple, β be any nondegenerate symmetric associative bilinear form on L . If $(x_1, \dots, x_n)$ is a basis of L , there is a uniquely determined dual basis $(y_1, \dots, y_n)$ relative to β s.t. $\beta(x_i, y_j) = \delta_{ij}$. If $x \in L$,

$$\begin{aligned}
a_{ik} &= \sum_j a_{ij} \beta(x_j, y_k) = \beta([xx_i], y_k) \quad \text{by the definition } [xx_i] = \sum_j a_{ij} x_j \\
&= \beta(-[x_i x], y_k) \quad \text{by the definition} \\
\left\{ \begin{array}{l} [xx_i] = \sum_j a_{ij} x_j \\ [xy_i] = \sum_j b_{ij} y_j \end{array} \right. & \text{Then} \quad = \beta(x_i, -[xy_k]) \quad \text{by the associativity of } \beta \\
&= -\sum_j b_{kj} \beta(x_i, y_j) \quad [xy_i] = \sum_j b_{ij} y_j \\
&= -b_{ki}
\end{aligned}$$

If $\phi : L \rightarrow \mathfrak{gl}(V)$ is any representation of L , write $c_\phi(\beta) = \sum_i \phi(x_i)\phi(y_i) \in \text{End} V$ where x_i, y_i run over dual bases relative to β . Then

$$\begin{aligned}
[\phi(x), c_\phi(\beta)] &= \sum_i [\phi(x), \phi(x_i)]\phi(y_i) + \sum_i \phi(x_i)[\phi(x), \phi(y_i)] \quad \text{by the definition} \\
&= \sum_{ij} a_{ij} \phi(x_j)\phi(y_i) + \sum_{ij} b_{ij} \phi(x_i)\phi(y_j) \quad \text{by the identity } [x, yz] = [xy]z + y[xz], \\
&= 0 \quad \text{by the identity } a_{ik} = -b_{ki}
\end{aligned}$$

i.e. $c_\phi(\beta)$ is an endomorphism of V commuting with $\phi(L)$.

Definition 1.2.5. Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be a faithful representation with trace form $\beta(x, y) = T(\phi(x)\phi(y))$. Having fixed a basis $(x_1, \dots, x_n)$ of L , we call $c_\phi = c_\phi(\beta)$ the Casimir element of ϕ .

$$\rightarrow \left\{ \begin{array}{l} \text{its trace is } \sum_i T(\phi(x_i)\phi(y_i)) = \sum_i \beta(x_i, y_i) = \dim L \\ \text{if } \phi \text{ is irreducible, by the Schur's lemma } c_\phi \text{ is scalar} \end{array} \right.$$

e.g. $L = \mathfrak{sl}(2, F)$ with the standard basis (x, h, y) , $V = F^2$, $\phi : L \rightarrow \mathfrak{gl}(V)$ the identity map. Then the dual basis relative to the trace form is $(y, h/2, x)$, so $c_\phi = xy + h^2/2 +$

$$yx = \begin{bmatrix} 3/2 & 0 \\ 0 & 3/2 \end{bmatrix} \text{ where } 3/2 = \dim L / \dim V.$$

When ϕ is not faithful, since $\ker \phi$ is an ideal of L , a sum of certain simple ideals. Let L' denote the sum of the remaining simple ideals by the theorem 1.2.4. Then the restriction of ϕ to L' is a faithful representation of L' and we can use the previous argument of faithful representation.

Lemma 1.2.5. *Let $\phi: L \rightarrow \mathfrak{gl}(V)$ be a representation of a semisimple Lie algebra L . Then $\phi(L) \subset \mathfrak{sl}(V)$. In particular, L acts trivially on any one dimensional L -module.*

pf. Since $L = [LL]$, given L semisimple, and $\mathfrak{sl}(V)$ is the derived algebra of $\mathfrak{gl}(V)$, the lemma follows.

Theorem 1.2.6 (Weyl). *Let $\phi: L \rightarrow \mathfrak{gl}(V)$ be a finite dimensional representation of a semisimple Lie algebra. Then ϕ is completely reducible.*

pf.

- V has an L -submodule W of codimension 1.

Since by the previous lemma L acts trivially on V/W , we may denote this module F and we get a s.e.s. $0 \rightarrow W \rightarrow V \rightarrow F \rightarrow 0$.

- We can reduce the case into the one where W is an irreducible L -module.

Let W' be a proper nonzero submodule of W . This yields a s.e.s $0 \rightarrow W/W' \rightarrow V/W' \rightarrow F \rightarrow 0$. By the inductive hypothesis, this splits, i.e. there exists a 1-dimensional L -submodule $\tilde{W}/W'$ of V/W' complementary to W/W' . So we have another s.e.s. $0 \rightarrow W' \rightarrow \tilde{W} \rightarrow F \rightarrow 0$ with $\dim W' < \dim W$. Thus by inductive hypothesis, $\tilde{W} = W' \oplus X$, for some submodule X complementary to W' . But $V/W' = W/W' \oplus \tilde{W}/W'$, and therefore $V = W \oplus X$ given
$$\begin{cases} \dim W + \dim X = \dim V \\ W \cap X = 0 \end{cases}.$$

So assume W is irreducible. Let $c = c_\phi$ be the Casimir element of ϕ . Since c commutes with $\phi(L)$, c is actually an L -module endomorphism of V and in particular $c(W) \subset W$ and $\ker c$ is an L -submodule of V . Because L acts trivially on V/W , c must do likewise. So c has trace 0 on V/W . On the other hand, c acts as a scalar on the irreducible L -submodule W by Schur's lemma, which cannot be 0 because that would force $T_V(c) = 0$, a contradiction. Thus $\ker c$ is a 1-dimensional L -submodule of V (given $c|_W$ invertible, $\ker c \cap W = 0$ and $\dim \ker c \geq \dim V - \dim W = 1$ and $\ker c \cap W = 0$, $\dim W = \dim V - 1$ forces

$\dim \ker c = 1$) which intersects W trivially. This is the desired complement to W .

- General case

Let W be a nonzero submodule of V . Then $\exists$ a s.e.s. $0 \rightarrow W \rightarrow V \rightarrow V/W \rightarrow 0$. Let $\text{hom}(V, W)$ be the space of linear maps $V \rightarrow W$, viewed as an L -module. Let $\mathfrak{V}$ be the subspace of $\text{hom}(V, W)$ consisting of those maps whose restriction to W is a scalar multiplication. Then $\forall f \in \mathfrak{V}$, $f|_W = a1_W$. So $\forall x \in L$, $\forall w \in W$, $(xf)(w) = xf(w) - f(xw) = a(xw) - a(xw) = 0 \rightarrow xf|_W = 0$, i.e. $\mathfrak{V}$ is an L -submodule. Let $\mathfrak{W}$ be the subspace of $\mathfrak{V}$ consisting of f whose restriction to W is zero. Then $\mathfrak{W}$ is also an L -submodule and L maps $\mathfrak{V}$ into $\mathfrak{W}$. Moreover $\mathfrak{V}/\mathfrak{W}$ has dimension 1, because each $f \in \mathfrak{V}$ is determined by the scalar $f|_W \bmod \mathfrak{W}$. Then we have a s.e.s. $0 \rightarrow \mathfrak{W} \rightarrow \mathfrak{V} \rightarrow F \rightarrow 0$ and we can use the previous case to get one dimensional submodule complementary to $\mathfrak{W}$. Let $f : V \rightarrow W$ span one dimensional complement to $\mathfrak{W}$ in $\mathfrak{V}$. Then by the previous lemma L acts trivially on that 1-dimensional submodule. So $xf = 0$, $\forall x \in L$ so f is an L -homomorphism. After multiplying by a nonzero scalar, we may assume $f|_W = 1_W$. Since f maps V into W and acts as 1_W on W , $V = W \oplus \ker f$.

Theorem 1.2.7. *Let $L \subset \mathfrak{gl}(V)$ be a semisimple linear Lie algebra. Then L contains the semisimple and nilpotent parts in $\mathfrak{gl}(V)$ of all its elements. In particular, the abstract and usual Jordan decomposition in L coincide.*

pf. Let $x \in L$ be arbitrary with Jordan decomposition $x = x_s + x_n$ in $\mathfrak{gl}(V)$. Since $\text{ad}x(L) \subset L$, $\begin{cases} \text{ad}x_s(L) \subset L \\ \text{ad}x_n(L) \subset L \end{cases}$ by the proposition 1.2.1., where $\text{ad} = \text{ad}_{\mathfrak{gl}(V)}$, i.e. $x_s, x_n \in N_{\mathfrak{gl}(V)}(L) = N$, which is a Lie subalgebra of $\mathfrak{gl}(V)$ including L as an ideal. Define $L_W = \{y \in \mathfrak{gl}(V) | y(W) \subset W, T(y|_W) = 0\}$. For example $L_V = \mathfrak{sl}(V)$. Since $L = [LL]$, $x|_W \in \mathfrak{sl}(W)$ so $T(x|_W) = 0$ and therefore L lies in all such L_W . Set $L' = N \cap (\cap_W L_W)$. Clearly, L' is a subalgebra of N including L as an ideal and moreover if $x \in L$, $x_s, x_n \in L_W$ and therefore in L' . By the Weyl's theorem, given L an ideal, $L' = L + M$ for some L -submodule M where the sum is direct. But $[LL'] \subset L$, given $L' \subset N$, so the action of L on M is trivial. Let W be any irreducible L -submodule of V . If $y \in M$, $[Ly] = 0$ so by the Schur's lemma, y acts on W as a scalar. On the other hand $T(y|_W) = 0$, given $y \in L_W$. Thus y acts on W as zero. V can be written as a direct sum of irreducible L -submodules by the Weyl's theorem, so $y = 0$. This implies $M = 0$, $L = L'$ and the theorem follows.

Corollary 1.2.6. *Let L be a semisimple Lie algebra $\phi : L \rightarrow \mathfrak{gl}(V)$ a representation of L . If $x = s + n$ is the abstract Jordan decomposition of $x \in L$, then $\phi(x) = \phi(s) + \phi(n)$ is the usual Jordan decomposition of $\phi(x)$.*

pf. ads is semisimple on L , so L is spanned by adx -eigenvectors. By applying ϕ , the algebra $\phi(L)$ is spanned by the eigenvectors of $\text{ad}_{\phi(L)}\phi(s)$, since L has this property relative to ads and therefore $\text{ad}_{\phi(L)}\phi(s)$ is semisimple. Similarly, $\text{ad}_{\phi(L)}\phi(n)$ is nilpotent and it commutes with $\text{ad}_{\phi(L)}\phi(s)$. Accordingly $\phi(x) = \phi(s) + \phi(n)$ is the abstract Jordan decomposition of $\phi(x)$ in the semisimple Lie algebra $\phi(L)$. Then by the previous theorem, the corollary follows.

1.3 Roots System

1.3.1 Root Space Decomposition

If L consisted entirely nilpotent elements, then L would be nilpotent by Engel's Theorem. Otherwise, we can find $x \in L$ s.t. whose semisimple part x_s in the abstract Jordan decomposition is nonzero, i.e. L possesses nonzero subalgebras consisting of semisimple elements. Such a subalgebra is called **toral**.

Lemma 1.3.1. *A toral subalgebra of L is abelian.*

pf. Let T be toral. Since adx is semisimple and F is algebraically closed, adx is diagonalisable. Suppose $[xy] = ax$, $a \neq 0$, i.e. $\text{ad}_T x$ has nonzero eigenvalue. Then $\text{ad}_T y(x) = -ay$ is itself an eigenvector of $\text{ad}_T y$ of eigenvalue 0. On the other hand, we can write x as a linear combination of eigenvectors of $\text{ad}_T y$, given y being semisimple. After applying $\text{ad}_T y$ to x , all that is left is a combination of eigenvectors which belong to nonzero eigenvalues if any, a contradiction.

Fix a maximal toral subalgebra H of L . For example, if $L = \mathfrak{sl}(n, F)$ has H , the set of diagonal matrices of trace 0. Since H is abelian by the previous lemma, $\text{ad}_L H$ is a commuting family of semisimple endomorphisms of L . Then $\text{ad}_L H$ is simultaneously diagonalisable, i.e. L is the direct sum of the subspaces $L_\alpha = \{x \in L \mid [hx] = \alpha(h)x, \forall h \in H\}$, where α ranges over H^* . The set of all nonzero $\alpha \in H^*$ for which $L_\alpha \neq 0$ is denoted by Φ , whose elements are called the **roots** of L relative to H .

Definition 1.3.1. *A root space decomposition or Cartan decomposition is defined as*

$$L = C_L(H) \oplus \coprod_{\alpha \in \Phi} L_\alpha$$

Proposition 1.3.1. $\forall \alpha, \beta \in H^*$, $[L_\alpha L_\beta] \subset L_{\alpha+\beta}$. If $x \in L_\alpha$, $\alpha \neq 0$, then adx is nilpotent. If $\alpha, \beta \in H^*$, $\alpha + \beta \neq 0$, then L_α is orthogonal to L_β relative to the killing form κ of L .

pf. $\forall \begin{cases} x \in L_\alpha \\ y \in L_\beta \end{cases}, h \in H$ implies $\text{adh}([xy]) = [[hx]y] + [x[hy]] = \alpha(h)[xy] + \beta(h)[xy] = (\alpha + \beta)(h)[xy]$ by the Jacobi identity. So the first stmt follows. Then the second stmt is the immediate consequence of the first stmt. Find $h \in H$ for which $(\alpha + \beta)(h) \neq 0$. Then if $x \in L_\alpha, y \in L_\beta$, $\begin{cases} \kappa([hx], y) = -\kappa([xh], y) = -\kappa(x, [hy]) \text{ by the associativity of } \kappa \\ \alpha(h)\kappa(x, y) = -\beta(h)\kappa(x, y) \\ (\alpha + \beta)(h)\kappa(x, y) = 0 \end{cases}$, which implies $\kappa(x, y) = 0$.

Corollary 1.3.1. *The restriction of the killing form to $L_0 = C_L(H)$ is nondegenerate.*

pf. By the theorem 1.2.3., κ is nondegenerate. On the other hand, L_0 is orthogonal to all L_α , $\alpha \in \Phi$ by the previous proposition. If $z \in L_0$ is orthogonal to L_0 as well, $\kappa(z, L) = 0 \rightarrow z = 0$.

Lemma 1.3.2. *If x, y are commuting endomorphisms of a finite dimensional vector space, with y nilpotent, then xy is nilpotent; in particular, $T(xy) = 0$.*

Proposition 1.3.2. *Let H be a maximal toral subalgebra of L . Then $H = C_L(H)$.*

pf. Write $C = C_L(H)$.

- **CLAIM : C contains the semisimple and nilpotent parts of its elements**

pf. $x \in C_L(H)$ so adx maps the subspace H of L into the subspace 0. Then by the previous proposition 1.2.1., $(\text{adx})_s, (\text{adx})_n$ have the same property. But $\begin{cases} (\text{adx})_s = \text{adx}_s \\ (\text{adx})_n = \text{adx}_n \end{cases}$, so the claim follows.

- **CLAIM : All semisimple elements of C lie in H .**

pf. If x is semisimple and centralises H , $H + Fx$ is toral. The sum of commuting semisimple elements is again semisimple and therefore by the maximality of H , $x \in H$.

- **CLAIM : The restriction of κ to H is nondegenerate.**

pf. Let $\kappa(h, H) = 0$, $h \in H$. If $x \in C$ is nilpotent, $\begin{cases} [xH] = 0 \\ \text{adx is nilpotent} \end{cases}$ imply $T(\text{adxady}) = 0$, $\forall y \in H$, or $\kappa(x, H) = 0$. Then by the previous claims, $\kappa(h, C) = 0$, and therefore $h = 0$ and the claim follows.

- **C is nilpotent.**

pf. If $x \in C$ is semisimple, $x \in H$ by the previous claim2 and $\text{ad}_C x = 0$ is certainly nilpotent. On the other hand, if $x \in C$ is nilpotent, then $\text{ad}_C x$ is a fortiori nilpotent. Now let $x \in C$ be arbitrary, $x = x_s + x_n$ and $x_s, x_n \in C$ by the previous claim1. Since $\text{ad}_C x$ is the sum of commuting nilpotents and is therefore itself nilpotent. Therefore by the Engel's theorem, C is nilpotent.

- Since κ is associative and $[HC] = 0$, $\kappa(H, [CC]) = 0$. If C is not abelian, $[CC] \neq 0$. Since C is nilpotent by the previous claim4, $Z(C) \cap [CC] \neq 0$. Let $0 \neq z \in Z(C) \cap [CC]$. Then by the previous claim2, z cannot be semisimple. Its nilpotent part n is therefore nonzero and lies in C by the previous claim 1, and therefore also lies in $Z(C)$ by the proposition 1.2.1. Then by the previous lemma, $\kappa(n, C) = 0$, a contradiction to the previous corollary. Thus C is abelian. If $C \neq H$, C contains a nonzero nilpotent element x by the previous claim 1,2. By the previous lemma, $\kappa(x, y) = T(\text{ad}_x \text{ad}_y) = 0$, $\forall y \in C$, a contradiction to the previous corollary, so $C = H$ and the proposition follows.

Corollary 1.3.2. *The restriction of κ to H is nondegenerate.*

$\rightarrow H \sim H^*$ in the sense that $\forall \phi \in H^*$, it corresponds to the unique element $t_\phi \in H$ satisfying $\phi(h) = \kappa(t_\phi, h)$, $\forall h \in H$.

Proposition 1.3.3. 1. Φ spans H^*

2. if $\alpha \in \Phi$, $-\alpha \in \Phi$

3. Let $\alpha \in \Phi$, $x \in L_\alpha$, $y \in L_{-\alpha}$. Then $[xy] = \kappa(x, y)t_\alpha$

4. If $\alpha \in \Phi$, $[L_\alpha L_{-\alpha}]$ is one dimensional with basis t_α

5. $\alpha(t_\alpha) = \kappa(t_\alpha, t_\alpha) \neq 0$, $\forall \alpha \in \Phi$

6. If $\alpha \in \Phi$ and x_α is any nonzero element of L_α , then there exists $y_\alpha \in L_{-\alpha}$ s.t. $x_\alpha, y_\alpha, h_\alpha = [x_\alpha y_\alpha]$ span a three dimensional simple subalgebra of L isomorphic to $\mathfrak{sl}(2, F)$ via $x_\alpha \mapsto \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $y_\alpha \mapsto \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $h_\alpha \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

7. $h_\alpha = \frac{2t_\alpha}{\kappa(t_\alpha, t_\alpha)}$, $h_\alpha = -h_{-\alpha}$

pf.

- If Φ fails to span H^* , there exists nonzero $h \in H$ s.t. $\alpha(h) = 0$, $\forall \alpha \in \Phi$ by the duality. Then $[h, L_\alpha] = 0$, $\forall \alpha \in \Phi$. Since $[hH] = 0$, $[hL] = 0$ or $h \in Z(L) = 0$, a contradiction. So the stnt follows.

- Let $\alpha \in \Phi$. If $-\alpha \notin \Phi$, $L_{-\alpha} = 0$, $\kappa(L_\alpha, L_\beta) = 0$, $\forall \beta \in H^*$ by the proposition 1.3.1. Thus $\kappa(L_\alpha, L) = 0$, a contradiction to the nondegeneracy of κ .
- Let $\alpha \in \Phi$, $x \in L_\alpha$, $y \in L_{-\alpha}$. Let $h \in H$ be arbitrary. Then $\kappa(h, [xy]) = \kappa([hx], y) = \alpha(h)\kappa(x, y) = \kappa(t_\alpha, h)\kappa(x, y) = \kappa(\kappa(x, y)t_\alpha, h) = \kappa(h, \kappa(x, y)t_\alpha)$, i.e. H is orthogonal to $[xy] - \kappa(x, y)t_\alpha$ and therefore $[xy] = \kappa(x, y)t_\alpha$ by the previous corollary.
- By 3, t_α spans $[L_\alpha L_{-\alpha}]$, provided $[L_\alpha L_{-\alpha}] \neq 0$. Let $0 \neq x \in L_\alpha$. If $\kappa(x, L_{-\alpha}) = 0$, then $\kappa(x, L) = 0$, a contradiction to the nondegeneracy of κ . Therefore $\exists 0 \neq y \in L_{-\alpha}$ for which $\kappa(x, y) \neq 0$. Then by 3, $[xy] \neq 0$.
- Suppose $\alpha(t_\alpha) = 0$ so that $[t_\alpha x] = 0 = [t_\alpha y]$, $\forall x \in L_\alpha$, $y \in L_{-\alpha}$. By 4, we can find such x, y satisfying $\kappa(x, y) \neq 0$. Modifying if necessary by a scalar, we may assume $\kappa(x, y) = 1$. Then $[xy] = t_\alpha$ by 3. Thus the subspace S of L spanned by x, y, t_α is a three dimensional solvable algebra, $S \cong \text{ad}_L S \subset \mathfrak{gl}(L)$. In particular, $\text{ad}_L S$ is nilpotent $\forall s \in [SS]$, so $\text{ad}_L t_\alpha$ is both semisimple and nilpotent, i.e. $\text{ad}_L t_\alpha = 0$. Thus $t_\alpha \in Z(L)$, a contradiction to the choice of t_α .
- Given $0 \neq x_\alpha \in L_\alpha$, find $y \in L_{-\alpha}$ s.t. $\kappa(x_\alpha, y_\alpha) = \frac{2}{\kappa(t_\alpha, t_\alpha)}$, which is guaranteed by 5 and the fact that $\kappa(x_\alpha, L_{-\alpha}) \neq 0$. Set $h_\alpha = 2t_\alpha/\kappa(t_\alpha, t_\alpha)$. Then $[x_\alpha y_\alpha] = h_\alpha$ by 3. Moreover $[h_\alpha x_\alpha] = \frac{2}{\alpha(t_\alpha)}[t_\alpha x_\alpha] = \frac{2\alpha(t_\alpha)}{\alpha(t_\alpha)}x_\alpha = 2x_\alpha$ and similarly, $[h_\alpha y_\alpha] = -2y_\alpha$. So $x_\alpha, y_\alpha, h_\alpha$ span a three dimensional subalgebra of L with the same multiplication table as $\mathfrak{sl}(2, F)$.
- t_α is defined by $\kappa(t_\alpha, h) = \alpha(h)$, $\forall h \in H$. Then $t_\alpha = -t_{-\alpha}$. Also by the definition of h_α , $h_{-\alpha} = \frac{2t_{-\alpha}}{\kappa(t_{-\alpha}, t_{-\alpha})} = \frac{-2t_\alpha}{\kappa(t_\alpha, t_\alpha)} = -h_\alpha$.

Proposition 1.3.4. 1. $\alpha \in \Phi$ implies $\dim L_\alpha = 1$. In particular, $S_\alpha = L_\alpha + L_{-\alpha} + H_\alpha$ and for given nonzero $x_\alpha \in L_\alpha$, there exists a unique $y_\alpha \in L_{-\alpha}$ satisfying $[x_\alpha y_\alpha] = h_\alpha$

2. If $\alpha \in \Phi$, the only scalar multiples of α which are roots are $\pm\alpha$.

3. If $\alpha, \beta \in \Phi$, then $\begin{cases} \beta(h_\alpha) \in \mathbb{Z} \\ \beta - \beta(h_\alpha)\alpha \in \Phi \end{cases}$

4. If $\alpha, \beta, \alpha + \beta \in \Phi$, $[L_\alpha L_\beta] = L_{\alpha+\beta}$

5. Let $\alpha, \beta \in \Phi$, $\beta \neq \pm\alpha$. Let r, q be the largest integers for which $\beta - r\alpha, \beta + q\alpha$ are roots respectively. Then $\forall \beta + i\alpha \in \Phi$, $-r \leq i \leq q$ and $\beta(h_\alpha) = r - q$.

pf.

- 1,2

For each pair of roots $\alpha, -\alpha$, let $S_\alpha \cong \mathfrak{sl}(2, F)$ be a subalgebra of L constructed in the previous proposition. By the Weyl's theorem and the theorem 1.2., we have a complete description of all finite dimensional S_α -modules, in particular, we can describe $\text{ad}_L S_\alpha$. Fix $\alpha \in \Phi$. Consider the subspace M of L spanned by H along with all root spaces of the form $L_{c\alpha}$, $c \in F^*$. This is an S_α -submodule of L , thanks to the proposition 1.3.1. The weights of h_α on M are integers 0 and $2c = c\alpha(h_\alpha)$. In particular, all c occurring here must be integral multiples of $1/2$. Now S_α acts trivially on $\ker \alpha$, a subspace of codimension 1 in H complementary to Fh_α , while S_α is an irreducible S_α -submodule of M . Taken together, $\ker \alpha$ and S_α exhaust the occurrences of the weight 0 for h_α . Therefore, the only even weights occurring in M are $0, \pm 2$. Thus 2α is not a root, i.e. twice a root is never a root. But then $(1/2)\alpha$ cannot be a root either, so 1 cannot occur as a weight of h_α in M . Then $M = H + S_\alpha$. In particular, $\dim L_\alpha = 1$ so S_α is uniquely determined as the subalgebra of L generated by $L_\alpha, L_{-\alpha}$, and the only multiples of a root α which are roots are $\pm\alpha$.

- 3,5

Let $\beta \neq \pm\alpha$. Set $K = \sum_{i \in \mathbb{Z}} L_{\beta+i\alpha}$. Then each root space is one dimensional and no $\beta+i\alpha$ can equal 0, so K is an S_α -submodule of L , with one-dimensional weight spaces for the distinct integral weights $\beta(h_\alpha) + 2i$, $\beta+i\alpha \in \Phi$. Obviously, not both 0 and 1 can occur as weights of this form, so K is irreducible. The highest weight must be $\beta(h_\alpha) + 2q$ if q is the largest integer for which $\beta+q\alpha$ is a root. Similarly for the lowest integer r and the roots $\beta-r\alpha$. The weights on K form an arithmetic progression with difference 2, which implies the roots $\beta+i\alpha$ form a string, called the α -string through β , $\beta-r\alpha, \dots, \beta, \dots, \beta+q\alpha$. Then $(\beta-r\alpha)(h_\alpha) = -(\beta+q\alpha)(h_\alpha)$ or $\beta(h_\alpha) = r-q$.

- 4

If $\alpha, \beta, \alpha+\beta \in \Phi$, $\text{ad} L_\alpha$ maps L_β onto $L_{\alpha+\beta}$, i.e. $[L_\alpha L_\beta] = L_{\alpha+\beta}$.

Theorem 1.3.1. *Let L be a semisimple Lie algebra over the algebraically closed field F of characteristic 0, H be a maximal toral subalgebra, $\Phi \subset H^*$ be the set of roots of L relative to H and $L = H + \coprod_{\alpha \in \Phi} L_\alpha$ be the root space decomposition. Define E be the real vector space obtained by extending the base field from $\mathbb{Q}$ to $\mathbb{R}$, i.e. $E = \mathbb{R} \otimes_{\mathbb{Q}} E_{\mathbb{Q}}$ where $E_{\mathbb{Q}}$ is the $\mathbb{Q}$ -subspace of H^* spanned by all the roots. Then*

1. Φ spans E and 0 does not belong to Φ .

2. If $\alpha \in \Phi$, $-\alpha \in \Phi$, but no other scalar multiple of α is a root.

3. If $\alpha, \beta \in \Phi$, then $\beta - \frac{2(\beta, \alpha)}{(\alpha, \alpha)}\alpha \in \Phi$

4. If $\alpha, \beta \in \Phi$, $\frac{2(\beta, \alpha)}{(\alpha, \alpha)} \in \mathbb{Z}$.

1.3.2 Root Systems

Fix a euclidean space E , i.e. a finite dimensional vector space over $\mathbb{R}$ endowed with a p.d. symmetric bilinear form (α, β) . Geometrically, a reflection in E is an invertible linear transformation leaving pointwise fixed some hyperplane and sending any vector orthogonal to that hyperplane into its negative. Then a reflection is orthogonal, i.e. preserves the inner product on E . Any nonzero vector α determines a reflection σ_α , with reflecting hyperplane $P_\alpha = \{\beta \in E | (\beta, \alpha) = 0\}$.

$$\sigma_\alpha(\beta) = \beta - \frac{2(\beta, \alpha)}{(\alpha, \alpha)}\alpha \equiv \beta - 2 \frac{(\beta, \alpha)}{(\alpha, \alpha)}\alpha$$

Lemma 1.3.3. *Let Φ be a finite set which spans E . Suppose all reflections σ_α ($\alpha \in \Phi$) leave Φ invariant. If $\sigma \in GL(E)$ leaves Φ invariant, fixes pointwise a hyperplane P of E , and sends some nonzero $\alpha \in \Phi$ to its negative, then $\sigma = \sigma_\alpha$.*

pf. Let $\tau = \sigma\sigma_\alpha$. Then $\begin{cases} \tau(\Phi) = \Phi \\ \tau(\alpha) = \alpha \end{cases}$ and τ acts as the identity on the subspace $\mathbb{R}\alpha$ as well as on the quotient $E/\mathbb{R}\alpha$. So all eigenvalues of τ are 1, and the minimal polynomial of τ divides $(T-1)^l$, $l \equiv \dim E$. On the other hand, since Φ is finite, not all vectors $\beta, \tau(\beta), \dots, \tau^k(\beta)$, $k \geq \text{Card}\Phi$ can be distinct, so some power of τ fixes β . Choose k large enough so that τ^k fixes all $\beta \in \Phi$. Because Φ spans E , $\tau^k = 1$, so the minimal polynomial of τ divides $T^k - 1$. Combined with the previous argument, τ has minimal polynomial $T - 1 = \gcd(T^k - 1, (T - 1)^l)$, i.e. $\tau = 1$.

Definition 1.3.2. *A subset Φ of the euclidean space E is called a root system in E if*

the following axioms are satisfied;

$$\left\{ \begin{array}{l} \Phi \text{ is finite, spans } E, \text{ and does not contain } 0 \\ \text{if } \alpha \in \Phi, \text{ the only multiples of } \alpha \text{ in } \Phi \text{ are } \pm\alpha \\ \text{if } \alpha \in \Phi, \text{ the reflection } \sigma_\alpha \text{ leaves } \Phi \text{ invariant} \\ \text{if } \alpha, \beta \in \Phi, \langle \beta, \alpha \rangle \in \mathbb{Z} \end{array} \right.$$

Let Φ be a root system in E . Denote by $\mathfrak{W}$ the subgroup of $GL(E)$ generated by the reflections σ_α ($\alpha \in \Phi$). Then by the third axiom, $\mathfrak{W}$ permutes the set Φ , which by the first axiom is finite and spans E . Thus we can identify $\mathfrak{W}$ with a subgroup of the symmetric group on Φ , in particular, $\mathfrak{W}$ is finite. Then $\mathfrak{W}$ is called the **Weyl group of Φ** .

Lemma 1.3.4. *Let Φ be a root system in E , with Weyl group $\mathfrak{W}$. If $\sigma \in GL(E)$ leaves Φ invariant, then $\sigma\sigma_\alpha\sigma^{-1} = \sigma_{\sigma(\alpha)}$, $\forall \alpha \in \Phi$, $\langle \beta, \alpha \rangle = \langle \sigma(\beta), \sigma(\alpha) \rangle$, $\forall \alpha, \beta \in \Phi$.*

pf. $\sigma\sigma_\alpha\sigma^{-1}(\sigma(\beta)) = \sigma\sigma_\alpha(\beta) \in \Phi$, given $\sigma_\alpha(\beta) \in \Phi$. But this equals $\sigma(\beta - \langle \beta, \alpha \rangle \alpha) = \sigma(\beta) - \langle \beta, \alpha \rangle \sigma(\alpha)$. Since $\sigma(\beta)$ runs over Φ as β runs over Φ , we conclude $\sigma\sigma_\alpha\sigma^{-1}$ leaves Φ invariant, while fixing pointwise the hyperplane $\sigma(P_\alpha)$ and sending $\sigma(\alpha)$ to $-\sigma(\alpha)$. By the previous lemma, $\sigma\sigma_\alpha\sigma^{-1} = \sigma_{\sigma(\alpha)}$. Then comparing the equation above with the equation $\sigma_{\sigma(\alpha)}(\sigma(\beta)) = \sigma(\beta) - \langle \sigma(\beta), \sigma(\alpha) \rangle \sigma(\alpha)$, the second stmt follows.

Definition 1.3.3. *Call (Φ, E) , (Φ', E') isomorphic if there exists a vector space isomorphism $\phi: E \rightarrow E'$ sending Φ onto Φ' s.t. $\langle \phi(\beta), \phi(\alpha) \rangle = \langle \beta, \alpha \rangle$, $\forall \alpha, \beta \in \Phi$.*

$\rightarrow \sigma_{\phi(\alpha)}(\phi(\beta)) = \phi(\sigma_\alpha(\beta))$ and therefore an isomorphism of root systems induces a natural isomorphism $\sigma \mapsto \phi \circ \sigma \circ \phi^{-1}$ of Weyl groups.

CLAIM : $\mathfrak{W}$ is a normal subgroup of $\text{Aut}\Phi$, the group of all isomorphisms of Φ onto itself.

pf. By the previous lemma, an automorphism of Φ is the same thing as an automorphism of E leaving Φ invariant.

Definition 1.3.4. $\Phi^\vee = \{\alpha^\vee \mid \alpha \in \Phi\}$ is called the dual of Φ , where $\alpha^\vee = \frac{2\alpha}{(\alpha, \alpha)}$.

$\rightarrow$ it is a root system in E whose Weyl group is canonically isomorphic to $\mathfrak{W}$.

pf. The first and second axiom holds trivially by the definition of $\alpha^\vee$, the multiple of α . $\sigma_{\alpha^\vee}(\beta^\vee) = \beta^\vee - 2(\beta^\vee, \alpha^\vee)/(\alpha^\vee, \alpha^\vee)\alpha^\vee = \frac{2\beta}{(\beta, \beta)} - 2(\frac{2\beta}{(\beta, \beta)}, \frac{2\alpha}{(\alpha, \alpha)})/(\frac{2\alpha}{(\alpha, \alpha)}, \frac{2\alpha}{(\alpha, \alpha)})\frac{2\alpha}{(\alpha, \alpha)}$. If $\alpha^\vee, \beta^\vee \in \Phi^\vee$, where $\Phi \subset H^*$, by the theorem 1.3.1., $\langle \beta, \alpha \rangle \in \mathbb{Z}$ and the axiom 4 is satisfied. Thus it is a root system.

$\rightarrow \alpha$ corresponds to t_α while $\alpha^\vee$ corresponds to h_α under the killing form identification of H^* with H .

The fourth axiom in the definition limits the possible angles occurring between pairs of roots. Since $\|\alpha\|\|\beta\|\cos\theta = (\alpha, \beta)$, $\begin{cases} \langle \beta, \alpha \rangle = \frac{2(\beta, \alpha)}{(\alpha, \alpha)} = 2\frac{\|\beta\|}{\|\alpha\|}\cos\theta \\ \langle \alpha, \beta \rangle \langle \beta, \alpha \rangle = 4\cos^2\theta \geq 0 \end{cases}$. Here $0 \leq \cos^2\theta \leq 1$, and $\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle \in \mathbb{Z}$ by the 4th axiom and $4\cos^2\theta \leq 4$, $\cos^2 \leq 1 \Rightarrow \langle \alpha, \beta \rangle \langle \beta, \alpha \rangle \in \{0, 1, 2, 3, 4\}$. $4\cos^2\theta = 4$ iff $\alpha = \pm\beta$, so for nonproportional roots, $\in \{0, 1, 2, 3\}$.

Lemma 1.3.5. *Let α, β be nonproportional roots. If $(\alpha, \beta) > 0$, i.e. if the angle between α, β is strictly acute, then $\alpha - \beta$ is a root. If $(\alpha, \beta) < 0$, $\alpha + \beta$ is a root.*

pf. Since (α, β) is positive iff $\langle \alpha, \beta \rangle$ is, i.e. one or the other of $\langle \alpha, \beta \rangle, \langle \beta, \alpha \rangle$ equals 1. If $\langle \alpha, \beta \rangle = 1$, $\sigma_\beta(\alpha) = \alpha - \beta \in \Phi$ by the third axiom of the definition. Similarly, if $\langle \beta, \alpha \rangle = 1$, $\beta - \alpha \in \Phi$, hence $\sigma_{\beta-\alpha}(\beta - \alpha) = \alpha - \beta \in \Phi$ and the first stmt follows. Then by the first stmt applied to $-\beta$, the second stmt follows.

$\rightarrow \forall \alpha, \beta$ nonproportional roots. Then look at all roots of the form $\beta + i\alpha$, the α -string through β . Let $r, q \in \mathbb{Z}_+$ be the largest integers for which $\begin{cases} \beta - r\alpha \in \Phi \\ \beta + q\alpha \in \Phi \end{cases}$. If some $\beta + i\alpha \notin \Phi$, $(-r \leq i \leq q)$, $\exists p < s$ in this interval s.t. $\beta + p\alpha \in \Phi$, $\beta + (p+1)\alpha \notin \Phi$, $\beta + (s-1)\alpha \notin \Phi$, $\beta + s\alpha \in \Phi$. But then by the lemma, $\begin{cases} (\alpha, \beta + p\alpha) \geq 0 \\ (\alpha, \beta + s\alpha) \leq 0 \end{cases}$. Since $p < s$, $(\alpha, \alpha) > 0$, a contradiction. Thus the α -string through β is unbroken, from $\beta - r\alpha$ to $\beta + q\alpha$. Here σ_α just adds or subtracts a multiple of α to any root, so this string is invariant under σ_α . So $r - q = \langle \beta, \alpha \rangle$ so the string has $r + q + 1$ roots and by the fact that $r + q \leq 3$, the root strings are of length at most 4.

Chapter 2

Representation of Lie Algebra